USN | | | | | | | | | | [22EC524]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

IMAGE PROCESSING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			5 x2=10
1.	**a.**	Discuss Optical Illusions with suitable example.	L2	CO1	**02**
	b.	Apply identity and negative transformation on the image f(x,y) = {0, 10,50, 5 , 95,100, 110,150,200}.	L3	CO2	**02**
	c.	Illustrate histogram and normalized histogram for the given image 0 0 0 0 0 1 2 3 0 2 4 6	L3	CO3	**02**
	d.	Outline a OpenCV code snippet to apply the Canny edge detection algorithm to a lung CT scan image. Analyze how edge detection is critical for tumor localization in lung cancer detection.	L3	CO4	**02**
	e.	Explain the concept of Intersection over Union (IoU) along with the formula used to calculate IoU in the context of evaluating the performance of image segmentation algorithms.	L2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Discuss any 5 applications of Digital Image Processing.	L2	C01	**05**
		OR			
	b.	Illustrate the process of image sampling and quantization of digital image formulation.	L3	CO2	**05**
3.	**a.**	Illustrate any 5 geometrical spatial transforms.	L3	CO3	**05**
		OR			
	b.	Given a set of pixel intensity values from an image: I= {70,75,80,85,90,95,100,105,110,115} Show the steps to derive the centroids after the first iteration by performing K-means clustering to segment these intensities into two clusters, assume the following initial centroids C1=75 and C2=105.	L3	CO4	**05**
4.	**a.**	Evaluate the Performance of Median Filtering and Gaussian Filtering on CT Scan Images using OpenCV, assessing Image Quality with Peak Signal-to-Noise Ratio (PSNR)	L2	CO5	**05**
		OR			

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	b.	Describe the following: **i.** Digital Image **ii.** Digital Image Processing	L2	CO1	**05**
5.	**a.**	Explain the components of Digital Image Processing with block diagram.	L2	CO1	**05**
		OR			
	b.	Illustrate the histogram equalization for the image given below and also show the histograms before and after equalization. 0 0 0 1 1 1 1 1 1 1 2 2 2 2 3 3 3 3 3 0 0 0 7 7	L3	CO3	**05**
6.	**a.**	Illustrate Bit plane slicing for the image on the image f(x,y) = {0, 10,50, 100, 5 , 95,150,200, 110,150,190,210,175,210,255,110}	L3	CO3	**05**
		OR			
	b.	Illustrate the Watershed Segmentation Algorithm in Image Processing.	L2	CO4	**05**
7.	**a.**	Evaluate the Performance of SIFT and ORB Feature Detection Algorithms on CT Scan Images using OpenCV. Hint: Use the number of key points detected as the evaluation metric.	L3	CO5	**05**
		OR			
	b.	Let P and Q are pixels at co-ordinates (10,15) and (15,20) respectively. Determine which distance measure give minimum distance between them?	L3	CO2	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	With diagram discuss the process of image acquisition using **i.** Single Sensor. **ii.** Sensor Strip. **iii.** Sensor Array.	L2	CO1	**10**
		OR			
	b.	Demonstrate brightness adaptation and discrimination in Image Processing.	L3	CO2	**10**
9.	**a.**	Explain the fundamental steps in Digital Image Processing with block diagram.	L2	CO1	**10**
		OR			
	b.	Apply the A] Logarithmic transformation, B] Negative Transformation, C] Thresholding Transformation D] Grey Level Slicing **i.** $s=\{M, 90\leq\gamma\leq 1600, Otherwise$ **ii.** $s=\{M, 90\leq\gamma\leq 160\gamma, Otherwise$ on the image	L3	CO2	**10**

		f(x,y) = {0, 10,50, 100, 5 , 95,150,200, 110,150,190,210,175,210,255,100}			
10.	**a.**	Analyze the three basic types of functions used frequently for image enhancement with appropriate examples for each.	L3	CO3	**10**
		OR			
	b.	**i.** Demonstrate the working principle of Prototype Matching in image classification. **ii.** Illustrate Optimum (Bayes) Statistical Classifier.	L3	CO4	**10**
11.	**a.**	Illustrate the histogram specification on image shown in the figure below: 0 1 0 2 2 3 3 2 0 1 0 1 1 3 2 0 Having ri = zi = 0,1,2,3 Pr(ri) = 0.25 for i = 0,1,2,3 Pz(Z0) = 0, Pz(Z1) = 0.5, Pz(Z2) = 0.5, Pz(Z3) = 0	L3	CO3	**10**
		OR			
	b.	Illustrate the architecture and functioning of a Deep Convolutional Neural Network (DCNN).	L3	CO4	**10**
12.	**a.**	Recommend Key Metrics for comparing image filtering and image segmentation algorithms	L5	CO5	**10**
		OR			
	b.	**i.** Illustrate the two resampling Methods with appropriate examples for each. **ii.** Illustrate the three interpolation methods with appropriate examples.	L3	CO2	**10**
13.	**a.**	Evaluate the performance of K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) Classifiers on Lung CT scan image dataset using accuracy as evaluation metric	L3	CO5	**10**
		OR			
	b.	Discuss any five properties of the Discrete Fourier Transform (DFT).	L2	CO2	**10**